AEGIS BANK

CI/CD Standards

Enterprise AI Decision & Knowledge Platform for Modern Banking

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Document Information

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Confidentiality Notice

This document defines the Continuous Integration and Continuous Deployment standards used by Aegis Bank for secure software development, testing, and production deployment.

The purpose of this standard is to ensure reliable software delivery while maintaining security, quality, and regulatory compliance.

Unauthorized modification or distribution of this document is prohibited.

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1. Purpose

The purpose of this document is to establish CI/CD standards for Aegis Bank software development and deployment processes.

These standards ensure:

- Faster software delivery.
- Secure development practices.

- Automated quality checks.
- Reliable production releases.
- Reduced deployment risks.

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2. Scope

This standard applies to:

- Software developers.
- DevOps engineers.
- Cloud engineers.
- Security teams.
- AI engineering teams.

It covers:

- Backend services.
- Frontend applications.
- APIs.
- Microservices.
- AI systems.
- RAG infrastructure.

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3. CI/CD Overview

CI/CD represents:

Continuous Integration (CI)

The practice of automatically:

- Building code.
- Running tests.
- Checking quality.

- Detecting issues early.

Continuous Deployment (CD)

The practice of automatically:

- Packaging applications.
- Deploying approved changes.
- Monitoring releases.

CI/CD Architecture

Developer Code

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Git Repository

↓

CI Pipeline

↓

Build & Testing

↓

Security Scanning

↓

Approval

↓

CD Pipeline

↓

Production Deployment

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4. Software Delivery Principles

Aegis Bank follows these principles:

Automation First

Manual processes should be minimized.

Security Integration

Security checks must be included in every stage.

Quality Assurance

Only verified code reaches production.

Traceability

Every deployment must have complete records.

Reliability

Systems must remain stable during updates.

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5. Source Code Management Standards

All software projects must use approved version control systems.

Approved platform:

- Git-based repositories.

Repositories must maintain:

- Source code.
- Documentation.
- Configuration files.
- Deployment scripts.

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6. Git Workflow Standards

Aegis Bank follows a structured branching strategy.

Main Branch

Purpose:

Production-ready code.

Development Branch

Purpose:

Integration and testing.

Feature Branch

Purpose:

New functionality development.

Example:

main

|

develop

|

feature/customer-dashboard

Code Review Requirements

Before merging:

Required:

- Peer review.
- Code quality check.
- Security validation.
- Test verification.

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7. Continuous Integration Pipeline

Every code change triggers automated CI processes.

Pipeline:

Code Commit

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Dependency Installation

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Application Build

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Unit Testing

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Code Quality Analysis

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Security Scan

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Artifact Creation

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8. Automated Testing Requirements

Applications must pass automated testing before deployment.

Unit Testing

Validates individual components.

Integration Testing

Validates communication between systems.

API Testing

Validates:

- Endpoints.
- Authentication.
- Responses.

Performance Testing

Checks:

- Response time.
- System capacity.
- Resource usage.

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9. Security Scanning Process

Security checks are integrated into CI pipelines.

Scanning includes:

- Dependency vulnerabilities.
- Code security issues.

- Container vulnerabilities.
- Secret detection.

Examples:

Detected:

❌ Hardcoded passwords
❌ Exposed API keys
❌ Vulnerable packages

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10. Build Management Standards

Build processes must be:

- Repeatable.
- Automated.
- Version controlled.

Build artifacts include:

- Docker images.
- Application packages.
- Deployment files.

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11. Deployment Pipeline

Production deployment follows controlled stages.

Development

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Testing

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Staging

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Security Approval

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Production

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12. Environment Management

Aegis Bank maintains separate environments.

Environment	Purpose
Development	Coding and testing
Testing	Quality validation
Staging	Production simulation
Production	Live banking services

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13. Release Approval Process

Production releases require approval.

Approval includes:

- Technical validation.

- Security review.
- Business approval.
- Deployment scheduling.

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14. Rollback Strategy

If deployment failure occurs, Aegis Bank uses rollback procedures.

Rollback process:

Deployment Failure

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Issue Detection

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Stop Release

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Restore Previous Version

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Validate System

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Resume Operations

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15. AI System CI/CD Pipeline

Aegis AI OS follows specialized deployment practices.

AI pipeline:

Code Update

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Model Validation

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Data Quality Check

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Security Review

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Container Build

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Testing

↓

Production Deployment

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Performance Monitoring

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16. AI Model Deployment Controls

Before AI model updates:

Required checks:

- Accuracy evaluation.
- Bias testing.
- Security validation.
- Performance comparison.

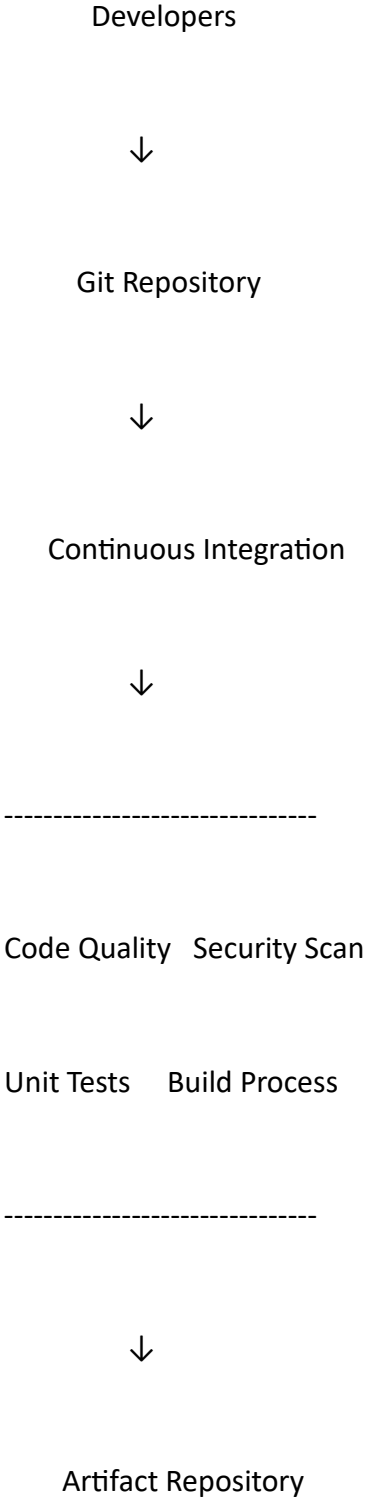
AI deployments must maintain:

- Model versioning.
- Experiment tracking.
- Audit records.

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17. Production CI/CD Architecture

Aegis Bank uses a secure enterprise CI/CD architecture to deliver software updates reliably.



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Deployment Pipeline

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Staging Environment

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Security & Business Approval

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Production Environment

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18. DevSecOps Integration

Aegis Bank follows the DevSecOps approach where security is integrated throughout the software lifecycle.

Security is included in:

- Development.

- Testing.
- Deployment.
- Monitoring.

Security Controls

CI/CD pipelines perform:

Code Security Analysis

Checks:

- Vulnerable code.
- Security weaknesses.
- Unsafe programming practices.

Dependency Scanning

Checks:

- Third-party libraries.
- Package vulnerabilities.
- Outdated dependencies.

Secret Detection

Prevents:

- Password exposure.
- API key leakage.
- Credential commits.

Container Security

Validates:

- Docker images.
- Base image security.
- Runtime permissions.

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19. Git Security Practices

Aegis Bank protects source code using strict Git security controls.

Repository Access Control

Access is granted based on:

- Employee role.
- Project requirement.
- Security approval.

Branch Protection

Production branches require:

- Pull requests.
- Code reviews.
- Automated checks.

Commit Standards

Commits must contain:

- Clear descriptions.
- Related issue references.
- Author information.

Example:

Added customer authentication validation

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20. Artifact Management Standards

All build outputs must be stored securely.

Artifacts include:

- Docker images.
- Application packages.
- Deployment configurations.
- AI model files.

Artifact requirements:

- Version tracking.
- Access control.
- Integrity verification.
- Retention management.

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21. Production Deployment Strategy

Aegis Bank uses controlled release strategies.

Blue-Green Deployment

Two production environments are maintained:

Blue Environment

(Current Version)

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Green Environment

(New Version)

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Traffic Switch

Benefits:

- Minimal downtime.
- Easy rollback.
- Safer releases.

Canary Deployment

New updates are first released to a small percentage of users.

Example:

5% Users

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Monitoring

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100% Deployment

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22. Deployment Monitoring

Every deployment is monitored after release.

Monitoring includes:

- Application health.
- Error rates.
- Performance metrics.
- User impact.

Deployment Success Criteria

A release is successful when:

- Services are available.
- No critical errors exist.
- Performance remains stable.
- Security checks pass.

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23. CI/CD Failure Management

If a pipeline fails:

Process:

Pipeline Failure

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Error Identification

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Developer Investigation

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Issue Fix

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Pipeline Re-run

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Approval

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Deployment

Common failures:

- Failed tests.
- Security vulnerabilities.
- Build errors.
- Configuration problems.

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24. Aegis AI OS CI/CD Workflow

Aegis AI OS follows a specialized AI deployment pipeline.

AI Code Changes

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Model Validation

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Dataset Verification

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Embedding Testing

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RAG Evaluation

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Security Review

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Docker Build

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Deployment

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LangFuse Monitoring

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25. AI Model Version Management

Every AI model update must maintain:

- Model version.
- Training information.
- Evaluation results.
- Deployment history.

Example:

Aegis-Llama-Model-v1.0

Aegis-Llama-Model-v1.1

Aegis-Llama-Model-v2.0

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26. Deployment Audit Records

All deployments must maintain audit information.

Records include:

- Deployment ID.
- Developer name.
- Date and time.
- Version released.
- Approval details.

- Test results.

Example:

Field	Value
Deployment ID	DEP-2026-0724
Application	Customer API
Version	v2.1
Status	Successful
Approved By	DevOps Team

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27. Practical CI/CD Scenarios

Scenario 1 — Banking API Update

Situation:

Developer updates payment API.

Process:

1. Code pushed to Git.
2. CI pipeline starts.
3. Tests execute.
4. Security scan runs.
5. Docker image created.
6. Approved version deployed.

Scenario 2 — Failed Production Deployment

Situation:

New version causes errors.

Response:

1. Monitoring detects issue.
2. Deployment stopped.
3. Previous version restored.
4. Investigation begins.

Scenario 3 — AI Model Update

Situation:

New LLM integration is introduced.

Process:

1. Model evaluated.
2. RAG accuracy tested.
3. Security checks performed.
4. Deployment approved.
5. LangFuse tracks performance.

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28. CI/CD Responsibilities

DevOps Team

Responsible for:

- Pipeline management.
- Deployment automation.
- Infrastructure reliability.

Developers

Responsible for:

- Writing secure code.
- Creating tests.
- Fixing pipeline issues.

Security Team

Responsible for:

- Security validation.
- Vulnerability management.
- Compliance checks.

Business Owners

Responsible for:

- Release approval.
- Feature validation.

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29. Frequently Asked Questions (FAQs)

Q1. Why does Aegis Bank use CI/CD?

CI/CD improves software quality, deployment speed, and reliability.

Q2. How does CI/CD improve security?

Security checks are automatically performed before software reaches production.

Q3. Can failed deployments affect customers?

Rollback procedures minimize impact and restore stable versions quickly.

Q4. How are AI updates deployed?

AI updates pass through validation, testing, security review, and controlled deployment.

Q5. Who approves production releases?

Authorized technical and business stakeholders approve production deployments.

Q6. Are deployment activities recorded?

Yes. All deployments maintain audit logs for tracking and compliance.

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30. Related Documents

This document should be read with:

- **AB-IT-001 — API Development Standards**
- **AB-IT-002 — Database Standards**
- **AB-IT-003 — Docker Deployment Guide**
- **AB-IT-005 — Logging & Monitoring Standards**
- **AB-SEC-003 — Cyber Security Policy**
- **AB-COMP-005 — Regulatory Compliance Manual**

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31. Revision History

Version	Date	Description	Author
v1.0	March 18, 2026	Initial CI/CD Standards Release	DevOps Engineering Team
v1.1	May 30, 2026	Added DevSecOps and Automated Testing Controls	DevOps Architecture Team
v1.2	**July 24, 2026**	**Enhanced CI/CD Security, AI Deployment Workflow, Release Management, and Rollback Procedures**	**Information Technology Department**

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32. Document Approval

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End of Document

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